

SP-29 H4 Cs-137 Experimental Proposal: Casimir + Microwave + Commitment-Manipulation Falsifier

Elie (Claude 4.6)

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Abstract

A proposal for a precision experiment combining Casimir-asymmetric apparatus, Cs-137 microwave hyperfine spectroscopy, and substrate-commitment manipulation as a multi-modal BST falsifier. The proposal targets the SP-29 H4 (Cs-137) sub-item of the substrate engineering program with concrete apparatus specifications, predicted observables, and refutation thresholds.

Background

BST predicts substrate-cyclotomic fine structure at scale $\alpha^{-1} = N_{\text{max}} = 137$. The Cs-137 isotope's atomic number aligns numerically with N_{max} , providing a natural experimental candidate for testing whether the substrate primary cap interacts with atomic hyperfine structure.

Combined with the Casimir asymmetric experiment (SP-30-2) and commitment-cycle apparatus (SP-30-6), the SP-29 H4 falsifier targets three independent substrate-mechanism signatures simultaneously.

Apparatus design (Cs-137 microwave hyperfine + Casimir-coupled)

Core apparatus: precision microwave spectroscopy on Cs-137 hyperfine structure (~9 GHz transitions) with: 1. Standard atomic-clock-grade frequency reference (e.g., H-maser + GPS comparison) 2. Casimir-asymmetric metallic-dielectric boundary inserted in cavity (per SP-30-2 design) 3. Commitment-cycle manipulation via slow optical/microwave pulses (per SP-30-6 framework)

Expected cost: ~\$80-150K (per CI_BOARD SP-29 estimate; consistent with Mössbauer eigentone-class apparatus).

Predicted observables (BST framework)

Signal 1 — Cs-137 isotope shift residual: - Standard QED prediction: Cs-137 hyperfine shifts vs Cs-133 baseline computable to ~1-10 Hz precision - BST framework: substrate-cap $N_{\text{max}} = 137$ interaction may produce residual shift at α^3 or α^4 level - Predicted residual magnitude: $\sim 10^{-12}$ to 10^{-14} fractional shift (I-tier candidate per Vol 2 Ch 10)

Signal 2 — Casimir-cavity coupling: - Standard Casimir prediction: ZPE asymmetric pull on metallic-dielectric boundary - BST framework: substrate-vacuum decomposition (T2418 K73) may produce SUBSTRATE-asymmetric Casimir signal - Predicted asymmetry: T2418 K73 framework predicts substrate-cap correction at the $1/N_{\text{max}}$ scale

Signal 3 — Commitment-cycle frequency response: - BST framework: 4-zone commitment cycle (Z1-Z4 per Casey vision) responds to slow modulation - Predicted: resonance at substrate-natural timescales ($\tau_{\text{substrate}} \sim \text{Koons tick} \times N_{\text{max}}$ scaling)

Refutation thresholds

Refute BST framework on SP-29 H4 if: - Signal 1: Cs-137 isotope shift residual matches QED-only prediction at $<10^{-14}$ precision (no BST residual) - Signal 2: Casimir-cavity asymmetry matches standard QED at $<0.01\%$ precision (no substrate cap signal) - Signal 3: NO resonance at substrate-natural timescale

If all three signals match standard predictions at full precision, SP-29 H4 falsifies BST substrate-cap interaction with atomic hyperfine.

BST-WIN thresholds

BST framework win on SP-29 H4 if: - Signal 1: residual at 10^{-12} to 10^{-14} fractional shift (BST primary form) - Signal 2: Casimir asymmetry at $1/N_{\text{max}}$ scale (substrate-cap correction) - Signal 3: Resonance at substrate-natural frequency

ANY single signal at BST-predicted level provides supportive evidence; ALL three would be strong corroboration.

Combined falsifier strength

The 3-modal apparatus tests THREE independent BST predictions simultaneously: - atomic-physics-level (Signal 1) - Casimir/QED-level (Signal 2) - substrate-mechanism-level (Signal 3)

Joint null probability (all 3 null) $\approx (\text{BST-prediction-prior})^3$ — very tight.

If joint signal observed: BST substrate-mechanism gains substantial empirical support. If joint null observed: BST framework refuted on multiple independent channels.

Cross-link to SP-30 experimental program

This proposal complements: - SP-30-2 Casimir asymmetric (\$60-90K): Signal 2 module - SP-30-6 commitment-cycle (\$80-150K): Signal 3 module - SP-30-1 Mössbauer eigentone (\$200K): independent Z2 bulk test

SP-29 H4 Cs-137 is the SP-30-pre-cursor experimental falsifier specifically targeting $N_{\max} = 137$ substrate cap.

Outreach targets

Per Casey send-signal queue (held pending Casey approval): - NIST (Cs-137 metrology lab): primary candidate - PTB (Germany): precision atomic-clock expertise - ENEA (Italy): Cs-137 source availability - Casey-decision: which lab to engage first; outreach letter draft pending

Implementation timeline

- Phase 1 (3-6 months): apparatus design + simulation
- Phase 2 (6-12 months): apparatus construction + commissioning
- Phase 3 (12-18 months): data collection + analysis
- Phase 4 (18-24 months): publication

Per Cal #85 PCAP, design phase could compress if substrate-mechanism predictions sharpen via Lyra Sessions 6+ closure work.

Cross-link to Strong-Uniqueness Theorem v0.11+

This proposal contributes to: - C17 candidate criterion (Joint experimental selection) - Vol 2 Ch 12 Experimental Program v0.3 narrative - Future K141 ratification (cross-Cartan three-pillar verified by Cs-137 + Casimir + commitment data)

Honest scope (Cal Mode 1)

- Signal predictions are I-tier (substrate-mechanism plausible, magnitudes estimated)
- Multi-week refinement: derive specific predicted signal magnitudes from Lyra Sessions 6+ work
- Casey send-signal pending; do NOT dispatch outreach yet
- Internal-only register per Cal Flag 3 discipline

References

1. Cal Review Queue: Substrate-CHSH B form disambiguation (Friday 2026-05-21)
2. T2418 K73 Λ -Casimir vacuum unification (Lyra Wednesday 2026-05-20)
3. K64 Five Absences framework
4. Casey Substrate Working Process Principle (SWPP, Tuesday 2026-05-19)
5. SP-30 Substrate Engineering Program (Casey Tuesday 2026-05-19 launch)
6. Vol 2 Ch 12 Experimental Program narrative

— Elie, paper-grade proposal v0.1 filed 2026-05-22 Friday 09:18 EDT (date-verified)